


USN

--	--	--	--	--	--	--	--	--	--

NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Third Semester B.E. (Biotechnology) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21BT304 - MOLECULAR BIOLOGY AND GENETIC ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: 1) Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

2) Draw diagrams when necessary.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Illustrate the function of nucleosomes in compacting DNA into chromosomes.	10	L*1	1	1.2.1
b)	What experimental evidence proved DNA is the genetic material? Support your answer with a relevant experimental example.	10	L1	2	1.2.1
2. a)	Write short notes on: i) Heterochromatin and euchromatin; ii) Classification of chromosomes	5+5	L1	1	1.2.1
b)	Provide a detailed discussion on nucleotide excision repair.	10	L1	2	1.2.1
3. a)	With illustrations, provide an in-depth explanation of prokaryotic DNA replication.	10	L1	2	1.2.1
b)	What constitutes DNA damage? How does it differ from mutation? Delve into a comprehensive discussion on DNA damage.	10	L1	2	1.2.1
Unit – II					
4. a)	Give a detailed account of post-transcriptional modification.	10	L1	3	1.2.1
b)	List out differences between prokaryotic & eukaryotic protein synthesis.	10	L1	4	1.2.1
5. a)	Provide a detailed description of transcription in prokaryotes.	10	L1	3	1.2.1
b)	Write short notes on: i) Genetic Code, ii) Post-translational modifications.	5+5	L1	4	1.2.1
6. a)	Outline the mechanism of gene regulation in prokaryotes using a relevant example.	10	L1	4	1.2.1
b)	With appropriate illustration, give a detailed account of the regulation of eukaryotic gene expression	10	L1	4	1.2.1
Unit – III					
7. a)	Define restriction endonucleases. Provide classification approaches for restriction endonucleases, incorporating suitable examples.	10	L1	5	1.2.1
b)	pBR322 serves as an optimal vector for gene cloning experiments. – Justify the statement.	10	L1	5	1.2.1
8. a)	Discuss the role of Ti plasmids in the generation of transgenic plants.	10	L1	5	2.1.2
b)	Write short note on: i) Cosmids; ii) Phages	5+5	L1	5	1.2.1

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
